



PENRITH SELECTIVE HIGH SCHOOL

**2017
HSC TRIAL EXAMINATION**

Mathematics Extension 2

General Instructions:

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A Reference Sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Answer all Questions in the writing booklets provided
- All diagrams are not drawn to scale

Total marks–100

SECTION I

Pages 3–7

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

SECTION II

Pages 8–15

90 marks

- Attempt Questions 11–16
- Allow about 2 hours 45 minutes for this section

Student Number: _____

This paper MUST NOT be removed from the examination room

Assessor: Mr Ferguson

Section I

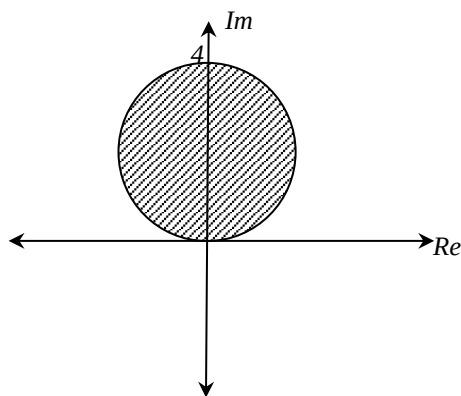
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Which of the following inequalities is represented by the Argand diagram?



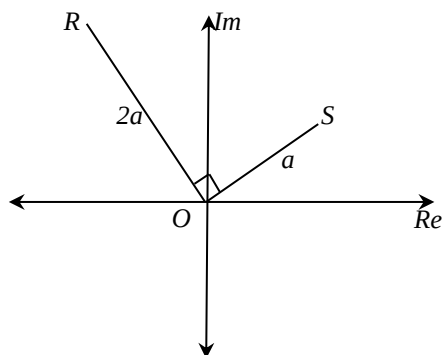
- (A) $|z - 2| \leq 2$
(B) $|z + 2| \leq 2$
(C) $|z - 2i| \leq 2$
(D) $|z + 2i| \leq 2$

- 2 Let $u = 7 \cos \frac{\pi}{4} + 7i \sin \frac{\pi}{4}$ and $v = a \cos b + ai \sin b$, where a and b are real constants.

If $uv = 42 \cos \frac{\pi}{20} + 42i \sin \frac{\pi}{20}$, then

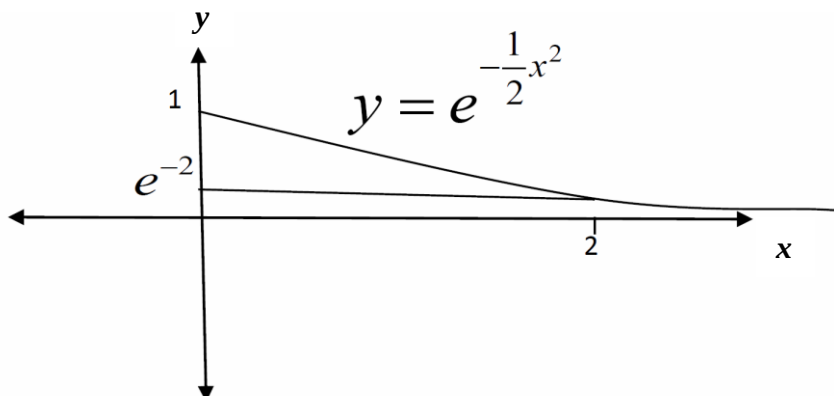
- (A) $a = 35$ and $b = \frac{\pi}{5}$
(B) $a = 6$ and $b = \frac{\pi}{5}$
(C) $a = 35$ and $b = -\frac{\pi}{5}$
(D) $a = 6$ and $b = -\frac{\pi}{5}$

- 3 In the Argand diagram below the points R and S represent the complex numbers w and z , respectively where $\angle SOR = 90^\circ$. The distance OR is $2a$ units, and distance OS is a units. Which of the following is correct?



- (A) $w = 2iz$
- (B) $w = i\bar{w}$
- (C) $w = \frac{-iz}{2}$
- (D) $w = \frac{-z}{2i}$
- 4 Which of the following graphs is the locus of the point P representing the complex number z moving in an Argand diagram such that $|z - 2i| = 2 + \text{Im } z$?
- (A) a circle
- (B) a parabola
- (C) a hyperbola
- (D) a straight line
- 5 The foci of the hyperbola $xy = 8$ are
- (A) $(\pm 4, \pm 4)$ (B) $(\pm 2\sqrt{2}, \pm 2\sqrt{2})$ (C) $(\pm 8\sqrt{2}, \pm 8\sqrt{2})$ (D) $(\pm 4\sqrt{2}, \pm 4\sqrt{2})$

- 6 The volume of the solid obtained by revolving the region bounded by $y = e^{-\frac{1}{2}x^2}$, $y = e^{-2}$ and the lines $x = 0, x = 2$ about the y – axis can be evaluated by which of the following integrals



- (A) $V = 2\pi \int_{e^{-2}}^1 x \left(e^{-\frac{1}{2}x^2} - e^{-2} \right) dx$
- (B) $V = 2\pi \int_{e^{-2}}^1 x \left(e^{-\frac{1}{2}x^2} \right) dx$
- (C) $V = 2\pi \int_0^2 x \left(e^{-\frac{1}{2}x^2} - e^{-2} \right) dx$
- (D) $V = 2\pi \int_0^2 x \left(e^{-\frac{1}{2}x^2} \right) dx$
- 7 Consider a polynomial $P(x)$ of degree 3.
You are given 2 numbers a and b such that

- $a < b$
- $P(a) > P(b) > 0$
- $P'(a) = P'(b) = 0$

The polynomial has

- (A) 3 real zeros
- (B) 1 real zero γ such that $\gamma < a$
- (C) 1 real zero γ such that $a < \gamma < b$
- (D) 1 real zero γ such that $\gamma > b$

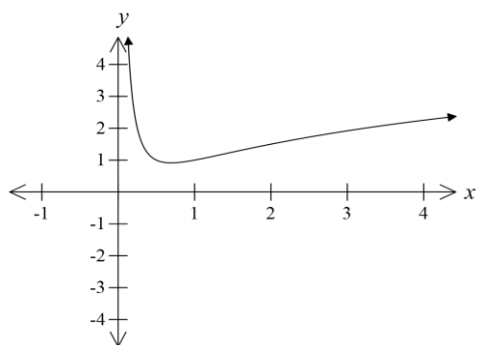
- 8 If $\int_1^4 f(x) dx = 6$, what is the value of $\int_1^4 f(5-x) dx$?
- (A) 6
- (B) 3
- (C) -1
- (D) -6

- 9 Given that $x^2 + y^2 + xy = 12$, which of the following is true?

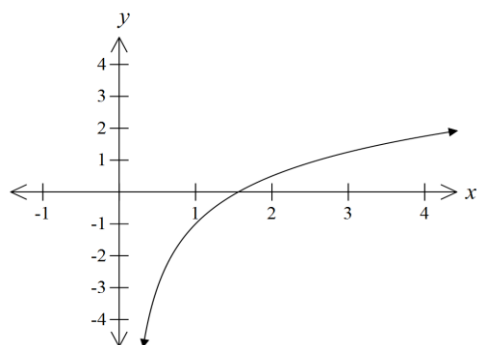
- (A) $\frac{dy}{dx} = \frac{2x+y}{2y+x}$
- (B) $\frac{dy}{dx} = -\frac{2x+y}{2y+x}$
- (C) $\frac{dy}{dx} = \frac{2x-y}{2y+x}$
- (D) $\frac{dy}{dx} = \frac{-2x+y}{2y+x}$

10 Which of the following is the sketch of $y = \log_2 x + \frac{1}{x}$?

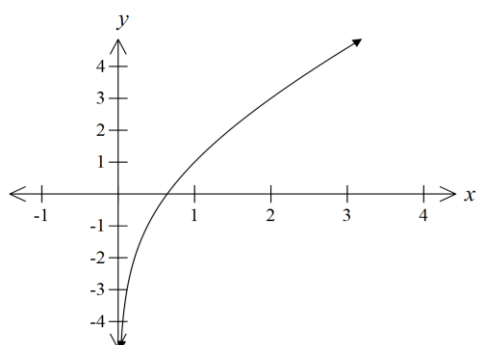
(A)



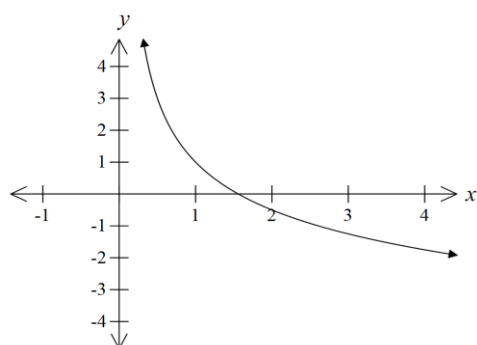
(B)



(C)



(D)



Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a new booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- a) Let $z = 3 + 2i$ and $w = -1 + i$. Express the following in the form $x + iy$, where x and y are real numbers:

i) $\frac{z}{iw}$ 2

ii) $\text{Im}(\bar{z}w)$ 2

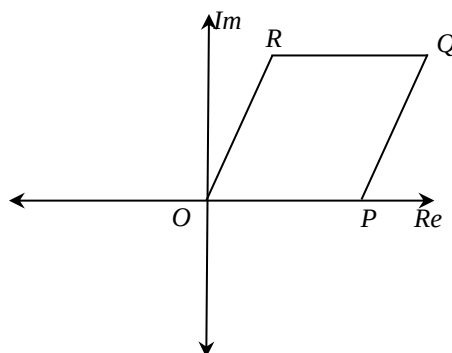
- b) Let $z = -1 + i$.

i) Express z in modulus-argument form. 2

ii) Express z^4 in modulus-argument form. 1

iii) Hence, evaluate z^{20} 1

- c) In the diagram below $OPQR$ is a rhombus. R represents $1 + i\sqrt{3}$. Find the complex number represented by Q 2



d) Evaluate $\int_0^1 \frac{2}{\sqrt{1+3x}} dx$ **2**

e) By using integration by parts, find $\int x^2 \ln 2x dx$ **3**

Question 12 (15 marks) Use a SEPARATE writing booklet.

- a) The polynomial $z^3 - 7z^2 + 25z - 39$ has one zero equal to $2 + 3i$. Write down its three linear factors. 2
- b) The equation $x^4 - px^3 + qx^2 - pqx + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Show that $(\alpha + \beta + \gamma)(\alpha + \beta + \delta)(\alpha + \gamma + \delta)(\beta + \gamma + \delta) = 1$ 4
- c) α, β, γ are the roots of $x^3 + qx + r = 0$. Find in terms of q, r the equation with roots $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$ 3
- d) Let ω be a non-real cube root of unity
- i) Show that $1 + \omega + \omega^2 = 0$ 1
- ii) Hence simplify $(1 + \omega)^2$ 1
- iii) Show that $(1 + \omega)^3 = -1$ 1
- iv) Use part iii) to simplify $(1 + \omega)^{3n}$ and hence show that
$${}^{3n}C_0 - \frac{1}{2}({}^{3n}C_1 + {}^{3n}C_2) + {}^{3n}C_3 - \frac{1}{2}({}^{3n}C_4 + {}^{3n}C_5) + {}^{3n}C_6 - \dots - {}^{3n}C_{3n} = (-1)^n$$
 3

Question 13 (15 marks) Use a SEPARATE writing booklet.

a) i) Prove that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ using the substitution $u = a - x$ 2

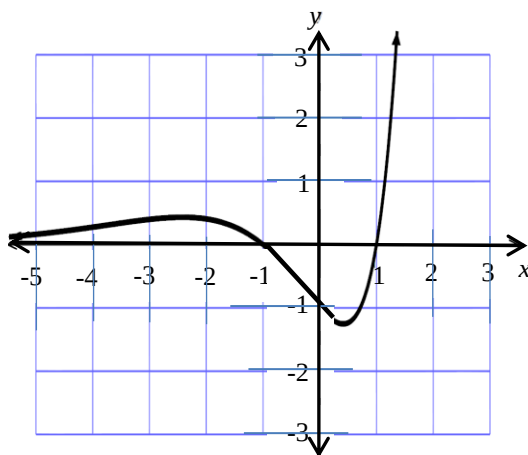
ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{1 + \sin x \cos x} dx$ 2

b) i) Find real constants A , B and C such that

$$\frac{4x-2}{(x^2-1)(x-2)} \equiv \frac{Ax+B}{(x^2-1)} + \frac{C}{(x-2)}$$
 3

ii) Hence evaluate $\int_3^6 \frac{4x-2}{(x^2-1)(x-2)} dx$ 2

c) The diagram shows the graph of $y = f(x)$



Draw separate $\frac{1}{3}$ page sketches of the following.

i) $|y| = f(x)$ 2

ii) $y = x \cdot f(x)$ 2

iii) $y = \sqrt{f(x)}$ 2

Question 14 (15 marks) Use a SEPARATE writing sheet.

- a) The equation $|z+4|+|z-4|=10$ corresponds to an ellipse in the Argand diagram.

Sketch the ellipse, and state the lengths of the major and minor axes

2

- b) Let $I_n = \int_0^{\pi} x^n \sin x \, dx$, where $n=0,1,2,\dots$

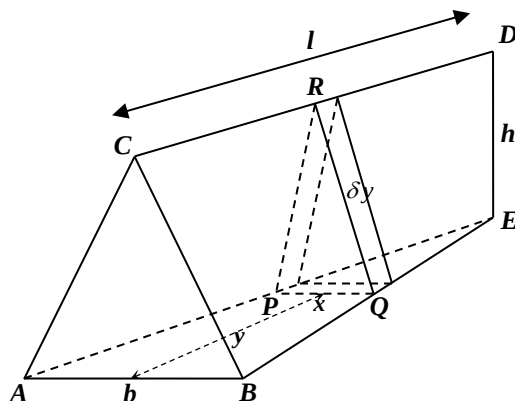
- i) Show that $I_n = \pi^n - n(n-1)I_{n-2}$ for $n=2,3,4,\dots$

3

- ii) Hence, evaluate $\int_0^{\pi} x^4 \sin x \, dx$

2

- c) ABC is an isosceles triangle with $AC=BC$ and $AB=b$. $ABCDE$ is a wedge shape with height $DE=h$ and length $CD=l$. Triangle ABC and line DE are perpendicular to the plane of ABE as shown in the diagram.



Consider a slice of the wedge height h and depth δy as in the diagram. The slice is parallel to the plane ABC at PQR .

- i) Show that the area of the triangle PQR can be expressed as $\frac{h}{2} \left(b - \frac{by}{l} \right)$.

2

- ii) Hence calculate the volume of the wedge

3

- d) Find $\int \frac{dx}{1+\sin x}$

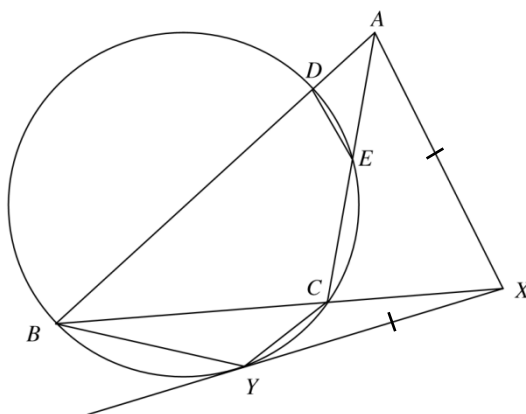
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Question 15 (15 marks) Use a SEPARATE writing booklet.

- a) Draw a large neat sketch of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ labelling clearly the, the foci, the directrices 3
- b) Find all the possible values of k if $\frac{x^2}{12-k} + \frac{y^2}{k+4} = 1$ represents an ellipse 2
- c) $P(a \sec \theta, b \tan \theta)$ is any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Given that the equation of the normal at P is given by $ax \tan \theta + by \sec \theta = (a^2 + b^2) \sec \theta \tan \theta$. (DO NOT PROVE THIS)
A line through P parallel to the y axis meets an asymptote at Q and the x axis at N .
The normal at P meets the x axis at R .
- i) Find the coordinates of Q, N, R . 3
- ii) Show that QR is perpendicular to the asymptote 2
- iii) Show that $OR = e^2 ON$ where e is the eccentricity. 2
- d) In a state swimming championships, 12 swimmers (including the Jones twins) are chosen to represent their club and are divided into three teams of four swimmers to form 3 relay teams.
- i) In how many ways can the 3 teams be formed? 2
- ii) Find the number of ways this can be done if the Jones twins (Angela and Bethany) are not to swim in the same relay team. 1

Question 16 (15 marks) Use a SEPARATE writing booklet.

- a) In the diagram XY is a tangent to the circle and $XY = XA$.



- i) Show that $\triangle XCY \parallel \triangle XBY$ 2
- ii) Hence explain why $\frac{XY}{BX} = \frac{CX}{XY}$ 1
- iii) Show that $\triangle AXC \parallel \triangle AXB$ 3
- iv) Prove that $DE \parallel AX$ 2

b) Given $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are positive real numbers. Where

$A_n = a_1 + a_2 + a_3 + \dots + a_n$ and $B_n = b_1 + b_2 + b_3 + \dots + b_n$, are such that $a_1, a_2, a_3, \dots, a_n > 0$, $b_1, b_2, b_3, \dots, b_n > 0$ and $A_r \leq B_r$, for $r = 1, 2, 3, \dots, n$.

(i) Prove, by mathematical induction for $n = 1, 2, 3, \dots$, that:

3

$$\begin{aligned} & \frac{1}{\sqrt{b_n}} B_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) B_{n-1} + \left(\frac{1}{\sqrt{b_{n-2}}} - \frac{1}{\sqrt{b_{n-1}}} \right) B_{n-2} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 \\ &= \sqrt{b_1} + \sqrt{b_2} + \sqrt{b_3} + \dots + \sqrt{b_n} \end{aligned}$$

(ii) Hence given

$$\begin{aligned} & \frac{a_1}{\sqrt{b_1}} + \frac{a_2}{\sqrt{b_2}} + \frac{a_3}{\sqrt{b_3}} + \dots + \frac{a_n}{\sqrt{b_n}} = \\ & \frac{1}{\sqrt{b_n}} A_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) A_{n-1} + \left(\frac{1}{\sqrt{b_{n-2}}} - \frac{1}{\sqrt{b_{n-1}}} \right) A_{n-2} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) A_1 \end{aligned}$$

$$\text{Show that } \sum_{r=1}^n \frac{a_r}{\sqrt{b_r}} \leq \sum_{r=1}^n \sqrt{b_r}$$

1

c) $\triangle ABC$ has sides a, b, c . If $a^2 + b^2 + c^2 = ab + bc + ca$, show that $\triangle ABC$ is equilateral.

3

End of Exam

Multiple choice.

Q1 C

2 D

3 A

4 $|x+iy-2i|=2+y$

$$|x+(y-2)i|=2+y$$

$$x^2+(y-2)^2=(2+y)^2$$

$$x^2+y^2-4y+4=4+4y+y^2$$

$$x^2=8y-4$$

$$x^2=4(2y-1)$$

$\therefore$ B (parabola).

5 $c=2\sqrt{2}$

$$\text{foci } (\pm 2\sqrt{2} \times \sqrt{2}, \pm 2\sqrt{2} \times \sqrt{2})$$

$$=(\pm 4, \pm 4)$$

$$= A.$$

6. C

7. B

8. A.

9. $2x+2y \frac{dy}{dx} + y + x \frac{dy}{dx} = 0$

$$(2y+x) \frac{dy}{dx} = -(2xy+y)$$

$$\frac{dy}{dx} = -\frac{(2x+1)y}{2y+x}$$

B

10. A.

$$\text{Q11 a) } z = 3 + 2i \quad w = -1 + i$$

$$\text{i) } \frac{z}{iw}$$

$$= \frac{3 + 2i}{i(-1 + i)}$$

$$= \frac{3 + 2i}{-i - 1} \times \frac{-1 + i}{-1 - i}$$

$$= \frac{-3 + 3i - 2i - 2}{(-1)^2 - i^2}$$

$$= \frac{-5 + i}{1 + 1}$$

$$= -\frac{5}{2} + \frac{1}{2}i$$

$$\text{ii) } \operatorname{Im}(\bar{z}w)$$

$$= \operatorname{Im}[(3 - 2i) \times (-1 + i)]$$

$$= \operatorname{Im}(-3 + 3i + 2i + 2)$$

$$= \operatorname{Im}(-1 + 5i)$$

$$= 5$$

Most students
have answered
part a) correctly.

Q11

$$b) z = -1 + i$$

$$i) |z| = \sqrt{(-1)^2 + 1^2}$$

$$|z| = \sqrt{2}$$

$$\arg(z) = \tan^{-1}\left(\frac{1}{-1}\right)$$

$$= \frac{3\pi}{4}$$

$$z = \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$ii) z^4 = \left[\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \right]^4$$

$$= (\sqrt{2})^4 \left(\cos \frac{3\pi}{4} \times 4 + i \sin \frac{3\pi}{4} \times 4 \right)$$

$$= 4 \left(\cos 3\pi + i \sin 3\pi \right)$$

$$= 4 \left(\cos \pi + i \sin \pi \right)$$

$$iii) z^{20} = (z^4)^5$$

$$= \left[4(\cos \pi + i \sin \pi) \right]^5$$

$$= 4^5 (\cos 5\pi + i \sin 5\pi)$$

$$= 1024 (\cos \pi + i \sin \pi)$$

$$= 1024 \times (-1)$$

$$= -1024$$

Some students obtained the wrong $\arg(z)$.

Some students didn't go on and evaluate z^{20} .

Q11
(1) R $1 + i\sqrt{3}$

$$|OR| = \sqrt{1^2 + 3^2} = 2$$

$$|OP| = |OR| \quad (\text{sides of a rhombus})$$

$$\begin{aligned}\vec{OQ} &= \vec{OR} + \vec{OP} \\ &= 1 + i\sqrt{3} + 2 \\ &= 3 + i\sqrt{3}\end{aligned}$$

$$\therefore Q(3 + i\sqrt{3})$$

d) $\int_0^1 \frac{2}{\sqrt{1+3x}} dx$

$$\text{Let } u = 1 + 3x$$

$$du = 3dx$$

$$= \frac{2}{3} \int_1^4 \frac{du}{\sqrt{u}}$$

$$x=1 \quad u=4$$

$$x=0 \quad u=1$$

$$= \frac{2}{3} \left[\frac{u^{1/2}}{1/2} \right]_1^4$$

$$= \frac{4}{3} (\sqrt{4} - \sqrt{1})$$

$$= \frac{4}{3}$$

e) $\int x^2 \ln 2x dx$

$$u = \ln 2x \quad v' = x^2$$

$$u' = \frac{1}{x}$$

$$v = \frac{x^3}{3}$$

$$= \frac{x^3}{3} \ln 2x - \int \frac{1}{x} \times \frac{x^3}{3} dx$$

$$= \frac{x^3}{3} \ln 2x - \frac{1}{3} \left[\frac{x^3}{3} \right] + C$$

$$= \frac{x^3}{3} \ln 2x - \frac{x^3}{9} + C$$

Students who understood vector representation did this question successfully.

$$\text{Q12. a) } P(z) = z^3 - 7z^2 - 25z - 39$$

$$P(2+3i) = 0$$

Since $2+3i$ is a root, then $2-3i$ is also a root for $P(z)$ (all real coefficients)

Let the roots of $P(z)$ be α, β, γ

$$\alpha + \beta + \gamma = 7$$

$$2+3i + 2-3i + \gamma = 7$$

$$\gamma = 3$$

$$\therefore P(z) = (z - (2+3i))(z - (2-3i))(z - 3)$$

$$\text{b) } x^4 - px^3 + qx^2 - pqx + 1 = 0$$

$$\alpha + \beta + \gamma + \delta = p$$

$$\alpha + \beta + \gamma = p - \delta$$

$$\alpha + \beta + \delta = p - \gamma$$

$$\alpha + \gamma + \delta = p - \beta$$

$$\beta + \gamma + \delta = p - \alpha$$

$$(p - \alpha)(p - \beta)(p - \gamma)(p - \delta)$$

$$= p^4 - (\sum \alpha)p^3 + (\sum \alpha\beta)p^2 - (\sum \alpha\beta\gamma)p + (-\alpha\beta\gamma\delta)$$

$$= p^4 - p \times p^3 + q \times p^2 - pq \times p + 1$$

$$= p^4 - p^4 + p^2q - p^2q + 1$$

$$= 1$$

most students could identify roots in terms of P , however there were struggles with expansion after.

Q12

$$c) x^3 + qx + r = 0$$

$$\text{let } x = \frac{1}{\alpha^2}$$

$$\alpha^2 = \frac{1}{x}$$

$$\alpha = \pm \frac{1}{\sqrt{x}}$$

$$\left(\pm \frac{1}{\sqrt{x}}\right)^3 + q\left(\pm \frac{1}{\sqrt{x}}\right) + r = 0$$

$$\pm \frac{1}{x\sqrt{x}} \pm \frac{q}{\sqrt{x}} + r = 0$$

$$\pm \frac{1}{\sqrt{x}} \left(\frac{1}{x} + q\right) = -r$$

$$\frac{1}{x} \left(\frac{1}{x} + q\right)^2 = (-r)^2$$

$$\frac{1}{x} \left(\frac{1+qx}{x}\right)^2 = r^2$$

$$\frac{(1+qx)^2}{x^3} = r^2$$

$$1 + 2qx + q^2x^2 = r^2x^3$$

$$\therefore r^2x^3 - q^2x^2 - 2qx - 1 = 0$$

Students forget
to square
to remove
surds.

Q12

d) i) If ω is a root of $z^3 = 1$

$$\omega^3 = 1$$

$$\omega^3 - 1 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

Since ω is a non-real root, $\omega \neq 1$.

$$\therefore \omega^2 + \omega + 1 = 0$$

$$\text{ii) } (1 + \omega)^2$$

$$= (-\omega^2)^2$$

$$= \omega^4$$

$$= \omega \times \omega^3$$

$$= \omega$$

$$\text{iii) } (1 + \omega)^3$$

$$= (-\omega^2)^3$$

$$= -\omega^6$$

$$= -(\omega^3)^2$$

$$= -1^2$$

$$= -1$$

$$\text{iv) } (1 + \omega)^{3n}$$

$$= [(1 + \omega)^3]^n$$

$$= (-1)^n \quad \text{from part iii)}$$

$$(1 + \omega)^{3n} = {}^{3n}C_0 + {}^{3n}C_1 \omega + {}^{3n}C_2 \omega^2 + {}^{3n}C_3 \omega^3 + \dots + {}^{3n}C_{3n} \omega^{3n}$$

$$(-1)^n = {}^{3n}C_0 + {}^{3n}C_1 \omega + {}^{3n}C_2 \omega^2 + {}^{3n}C_3 + \dots + {}^{3n}C_{3n}$$

$$\omega = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\omega^2 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$(-1)^n = {}^{3n}C_0 + {}^{3n}C_1 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) + {}^{3n}C_2 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) + {}^{3n}C_3 + \dots + {}^{3n}C_{3n}$$

Some students didn't explain why $\omega - 1 \neq 0$.

students were able to get $(1 + \omega)^{3n} = (-1)^n$,

however they struggled with the rest of this question.

$$(-1)^n = {}^{3n}C_0 + \left(-\frac{1}{2}\right)({}^{3n}C_1 + {}^{3n}C_2) + \frac{\sqrt{3}}{2}i({}^{3n}C_1 - {}^{3n}C_2) \\ + {}^{3n}C_3 + \dots + {}^{3n}C_{3n}$$

Equate real and imaginary part.

$$\times \frac{\sqrt{3}}{2}i({}^{3n}C_1 - {}^{3n}C_2) + \frac{\sqrt{3}}{2}i({}^{3n}C_4 - {}^{3n}C_5) + \dots = 0$$

$$\times {}^{3n}C_0 - \frac{1}{2}({}^{3n}C_1 + {}^{3n}C_2) + {}^{3n}C_3 - \frac{1}{2}({}^{3n}C_4 + {}^{3n}C_5) \\ + {}^{3n}C_6 - \dots + {}^{3n}C_{3n} = (-1)^n$$

Q13. a)

$$\begin{aligned}
 \text{i) } \int_0^a f(x) dx & \quad \text{let } u = a - x \\
 & \quad du = -dx \\
 & \quad x = a \quad u = 0 \\
 & \quad x = 0 \quad u = a \\
 & = \int_a^0 f(u) \times (-du) \\
 & = \int_0^a f(u) du \\
 & = \int_0^a f(x) dx
 \end{aligned}$$

$$\text{ii) } I = \int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{1 + \sin x \cos x} dx$$

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \frac{\cos(\frac{\pi}{2} - x) - \sin(\frac{\pi}{2} - x)}{1 + \sin(\frac{\pi}{2} - x) \cos(\frac{\pi}{2} - x)} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{1 + \cos x \sin x} dx
 \end{aligned}$$

$$2I = 0$$

$$I = 0$$

$$b) i) \frac{4x-2}{(x^2-1)(x-2)} = \frac{Ax+B}{x^2-1} + \frac{C}{x-2}$$

$$(Ax+B)(x-2) + C(x^2-1) = 4x-2$$

$$\text{let } x=2 \quad 3C=6 \\ C=2$$

$$\text{let } x=0 \quad -2B-C=-2 \\ -2B=0 \\ B=0$$

$$\text{let } x=1 \quad (A+B)x(-1)+0=2 \\ A=2$$

$$\frac{4x-2}{(x^2-1)(x-2)} = \frac{-2x}{x^2-1} + \frac{2}{x-2}$$

$$ii) \int_3^6 \frac{4x-2}{(x^2-1)(x-2)} dx$$

$$= \int_3^6 \left(\frac{-2x}{x^2-1} + \frac{2}{x-2} \right) dx$$

$$= \left[-\ln|x^2-1| + 2\ln|x-2| \right]_3^6$$

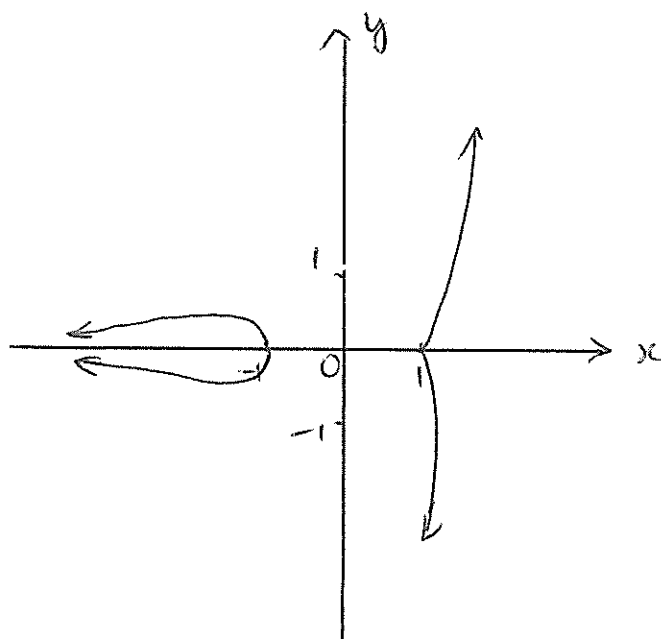
$$= \left[-\ln 35 - (-\ln 8) + 2\ln 4 - 2\ln 1 \right]$$

$$= -\ln 35 + \ln 8 + \ln 16$$

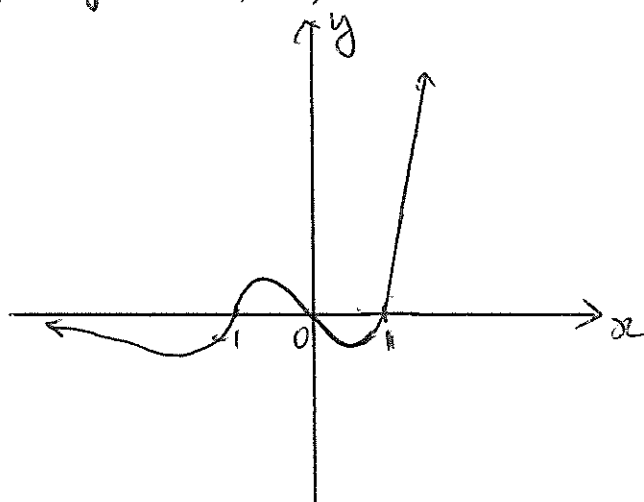
$$= \ln \left(\frac{16 \times 8}{35} \right)$$

$$= \ln \left(\frac{128}{35} \right)$$

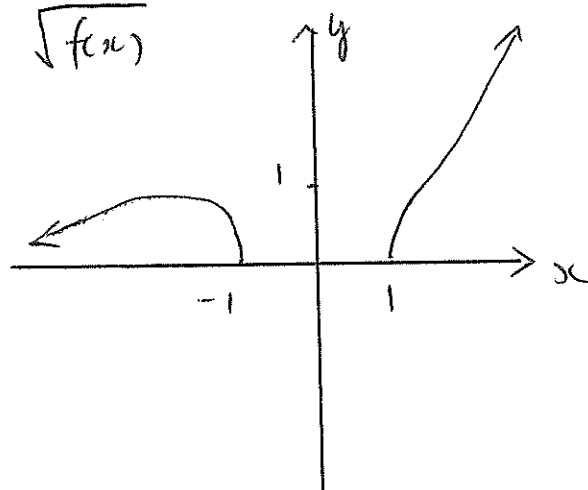
c) i) $|y| = f(x)$

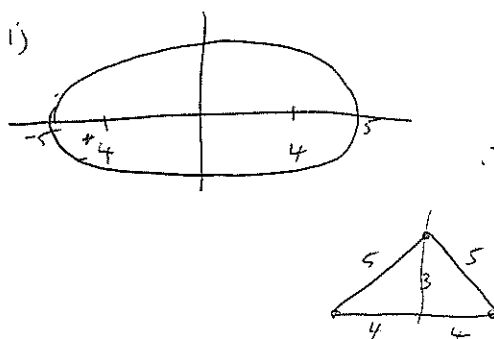


ii) $y = x \cdot f(x)$

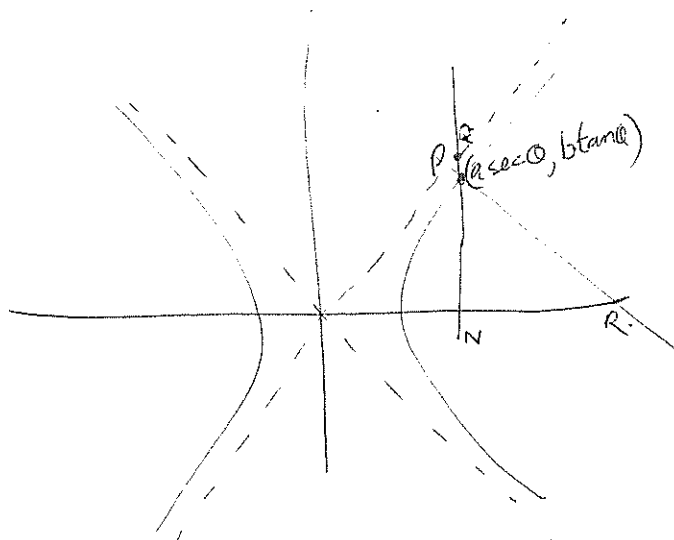


iii) $y = \sqrt{f(x)}$



Exam	MATHEMATICS : Question.....14	
Suggested Solutions		Marker's Comments
Question 14: a(i)  $PS + PS' = 2a$ $\therefore$ Major axis is 10 $\therefore b = 3$ $\therefore$ minor axis is 6 b. (i) $\int_0^\pi x^n \sin x \, dx$ $u = x^n \quad v' = \sin x$ $u' = nx^{n-1} \quad v = -\cos x$ $I_n = -x^n \cos x + n \int_0^\pi x^{n-1} \cos x \, dx$ $u = x^{n-1} \quad v' = \cos x$ $u' = (n-1)x^{n-2} \quad v = \sin x$ $I_n = [-x^n \cos x]_0^\pi + n [-x^{n-1} \sin x]_0^\pi - n \int_0^\pi (n-1)x^{n-2} \sin x \, dx$ $I_n = (-\pi^n \cos \pi - 0) + n(\pi^{n-1} \sin \pi - 0) - n(n-1)I_{n-2}$ $I_n = -\pi^n \times (-1) - n(n-1)I_{n-2}$ $\therefore I_n = \pi^n - n(n-1)I_{n-2}$ (ii) $\int_0^{\frac{\pi}{2}} x^4 \sin x \, dx$ $I_4 = \pi^4 - 4 \times 3 \times I_2$ $I_2 = \pi^2 - 2 \times 1 \times I_0$ $I_0 = \pi^0 - 0 \times I_{-2}$ $I_0 = \pi^0 - 0 = 1$ $I_4 = \pi^4 - 12 \times (\pi^2 - 4)$ $I_4 = \pi^4 - 12\pi^2 + 48$ $I_0 = \int_0^{\frac{\pi}{2}} x^0 \sin x \, dx$ $= [-\cos x]_0^{\frac{\pi}{2}} = 1$		well done - some students gave a semi axis measurement well done - well done

(c)

(1) N is $(a \sec \theta, 0)$ asymptote $y = \pm \frac{b}{a}x$ $Q(a \sec \theta, \frac{b}{a} \sec \theta)$ $\therefore Q(a \sec \theta, b \sec \theta)$ for R $a x \tan \theta + b y \sec \theta = (a^2 + b^2) \sec \theta \tan \theta$ when $y=0$, $a x \tan \theta = (a^2 + b^2) \sec \theta \tan \theta$

$$x = \frac{(a^2 + b^2) \sec \theta}{a}$$

 $R\left(\frac{(a^2 + b^2) \sec \theta}{a}, 0\right)$

students forgot
the $\pm$ case.
This wasn't
penalised

Exam	MATHEMATICS : Question. 14	
Suggested Solutions		Marker's Comments
(ii) asymptote $y = \pm \frac{b}{a}x$ $m = \pm \frac{b}{a}$ $M_{QR} = \frac{b \sec \theta - 0}{a \sec \theta - \frac{(a^2 + b^2) \sec \theta}{a}}$ $= \frac{ab \sec \theta}{a^2 \sec \theta - a^2 \sec \theta - b^2 \sec \theta}$ $= -\frac{a}{b}$ $M_{OR} = \frac{-b \sec \theta}{a \sec \theta - \frac{(a^2 + b^2) \sec \theta}{a}}$ $= \frac{a}{b}$ $-\frac{a}{b} \times \frac{b}{a} = -1 \quad \frac{a}{b} \times -\frac{b}{a} = -1$ $\therefore QR \perp$ asymptote (iii) $OR = \frac{(a^2 + b^2) \sec \theta}{a}$ $ON = a \sec \theta$ $b^2 = a^2(e^2 - 1)$ $\frac{b^2}{a^2} + 1 = e^2$ $\frac{b^2 + a^2}{a^2} = e^2$ $e^2 ON = \frac{b^2 + a^2}{a^2} \times a \sec \theta$ $= \frac{(b^2 + a^2) \sec \theta}{a}$ $= OR$ d) (i) $\frac{{}^{12}C_4 \cdot {}^8C_4 \cdot {}^4C_4}{3!}$ (ii) ${}^{10}C_3 \cdot {}^7C_3 \cdot {}^4C_4$ $= 4200$		$M_1, M_2 = -1$ concept needed to be used at some stage. some students made algebraic errors. well done. (ii) some students used correct alternative method from part (i). working was essential

Question 15

$$a) \frac{x^2}{25} + \frac{y^2}{16} = 1$$

foci $(\pm ae, 0)$ Directrices $x = \pm \frac{a}{e}$.

$$b^2 = a^2(1 - e^2)$$

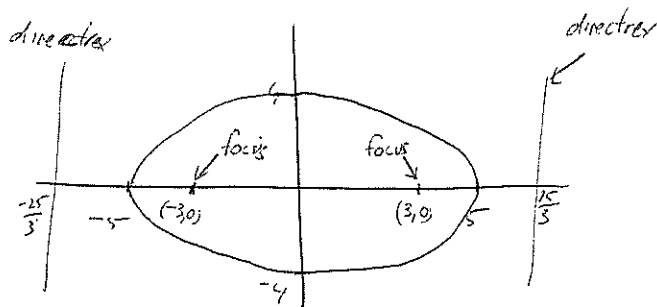
$$16 = 25(1 - e^2)$$

$$\frac{16}{25} = (1 - e^2)$$

$$e^2 = \frac{9}{25}$$

$$e = \frac{3}{5}$$

foci $(\pm 3, 0)$ Directrices $x = \pm \frac{25}{3}$



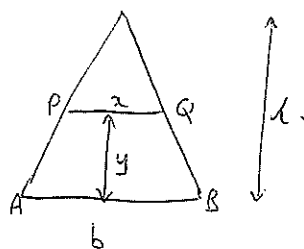
$$b) \begin{aligned} 12 - k &> 0 & k + 4 &> 0 \\ k &< 12 & \text{and} & k > -4 \\ -4 &< k < 12 \end{aligned}$$

well done

students used
more complicated
approaches
and
forgot $a^2 = (12 - k)$
etc
and ended
up with
complicated
algebraic
expressions

C(i)

$$\begin{aligned}
 A_{\Delta PQR} &= \frac{1}{2} \times PQ \times h \\
 &= \frac{1}{2} x h \\
 &= \frac{1}{2} \left(b - \frac{b}{l} y\right) h
 \end{aligned}$$



$$\begin{aligned}
 \frac{x}{b} &= \frac{l-y}{l} \\
 x &= \frac{b(l-y)}{l} \\
 x &= b - \frac{by}{l}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } V &= \int_0^l \frac{1}{2} \left(b - \frac{b}{l} y\right) h \times dy \\
 &= \int_0^l \left(\frac{1}{2} bh - \frac{1}{2} \frac{bh}{l} y\right) dy \\
 &= \left[\frac{1}{2} bhy - \frac{bh}{l} \cdot \frac{y^2}{4} \right]_0^l \\
 &= \frac{1}{2} bh \times l - \frac{bh}{4l} \times l^2 \\
 &= \frac{1}{2} bhl - \frac{bhl}{4} \\
 &= \frac{1}{4} bhl
 \end{aligned}$$

$$\text{d) } \int \frac{dx}{1+\sin x}$$

$$\begin{aligned}
 &= \int \frac{1}{1+\frac{2t}{1+t^2}} \times \frac{2 dt}{1+t^2} \\
 &= \int \frac{1+t^2}{1+t^2+2t} \times \frac{2 dt}{1+t^2} \\
 &= 2 \int \frac{dt}{t^2+2t+1}
 \end{aligned}$$

$$= 2 \int \frac{dt}{(t+1)^2} = 2 \frac{(t+1)^{-1}}{-1} + C = \frac{-2}{t+1} + C = \frac{-2}{\tan \frac{x}{2} + 1} + C$$

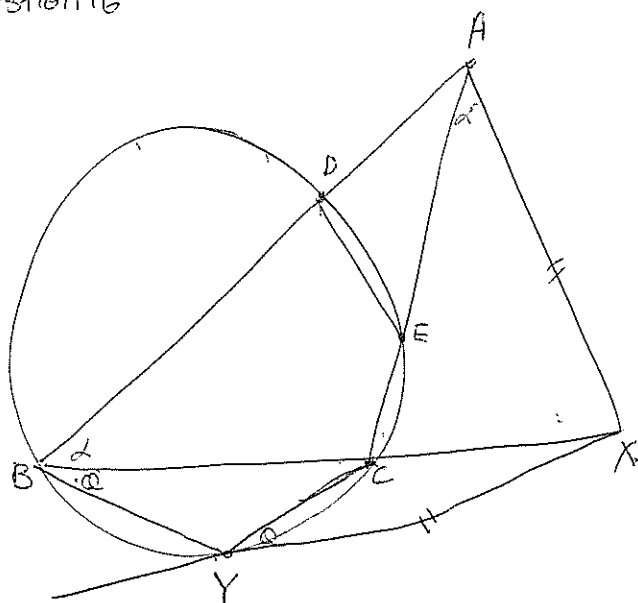
$$\begin{aligned}
 \text{Let } t &= \tan \frac{x}{2} \\
 dt &= \frac{1}{2} \sec^2 \frac{x}{2} dx \\
 2dt &= (1+t^2) dx \\
 \frac{2dt}{(1+t^2)} &= dx
 \end{aligned}$$

some students
lacked understanding
and $\int_0^h$ etc.
this was
penalised

well done.
Students
need to remember
to convert
back to $\tan \frac{x}{2}$

Question 16

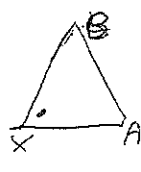
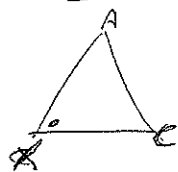
a)



- (i) In $\triangle XCY$ and $\triangle XBY$
 $\angle BXY = \angle CXY$ (common)
 $\angle CYX = \angle XBY$ (alternate segment theorem)
 $\therefore \triangle XCY \parallel \triangle XBY$ (equiangular)

(ii) $\frac{XY}{CX} = \frac{BY}{CY} = \frac{BX}{XY}$ (corresponding sides of similar triangles are in proportion)
 $\therefore \frac{XY}{BX} = \frac{CX}{XY}$

- (iii) In $\triangle AXC$ and $\triangle AXB$
 $\angle AXC = \angle AXC$ (common)



$XC \cdot XB = XY^2$ (tangent and secant)
 $\therefore XC \cdot XB = AX^2$ ($XY = AX$ given)
 $\therefore \frac{AX}{XB} = \frac{XC}{AX}$
 $\therefore \triangle AXC \parallel \triangle AXB$ (included angle equal with corresponding sides in equal ratios)

well done

well done

Some students used different methods but most were successful.

Exam	MATHEMATICS : Question 16	
Suggested Solutions		Marker's Comments
(IV) $DE \parallel AX$. Let $\angle ABX = \alpha$. $\angle XAC = \angle XBA$ (from part iii - corresponding angles are equal) $\therefore \angle XAC = \alpha$. $\angle CED = 180 - \alpha$ (opposite angles in a cyclic quadrilateral are supplementary) $\therefore \angle AED = \alpha$ (straight angle) So $\angle XAC = \angle AED$ and since the angles are alternate and equal then $DE \parallel AX$. b(i) Test $n=1$ $A_1 = a$, $B_1 = b$, $\frac{1}{\sqrt{b_1} B_1} = \frac{1}{\sqrt{b_1} \times b_1}$ $= \frac{1}{\sqrt{b_1}}$: proven true for $n=1$ $= R.H.S$ Assume true for $n=k$ $\frac{1}{\sqrt{b_k}} B_k + \left(\frac{1}{\sqrt{b_{k+1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k+1} + \dots + \frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} B_1 = \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_k}$ prove true for $n=k+1$ R.T.P $\frac{1}{\sqrt{b_{k+1}}} B_{k+1} + \left(\frac{1}{\sqrt{b_k}} - \frac{1}{\sqrt{b_{k+1}}} \right) B_k + \left(\frac{1}{\sqrt{b_{k+1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k+1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1$ $= \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_k} + \sqrt{b_{k+1}}$ $\frac{1}{\sqrt{b_{k+1}}} B_{k+1} - \frac{1}{\sqrt{b_{k+1}}} B_k + \frac{1}{\sqrt{b_k}} B_k + \left(\frac{1}{\sqrt{b_{k+1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k+1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_k}} \right) B_1$ $\frac{1}{\sqrt{b_{k+1}}} [B_{k+1} - B_k] + \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_k}$ (from assumption) now $B_{k+1} - B_k = (b_1 b_2 \dots b_{k+1}) - (b_1 b_2 \dots b_k)$ $= b_{k+1}$ So $\frac{1}{\sqrt{b_{k+1}}} \cdot b_{k+1} + \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_k} = \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_k} + \sqrt{b_{k+1}}$		well done. students generally tested for $n=1$ and assumed for $n=k$ correctly. for $n=k+1$ unless students saw $B_{k+1} - B_k = b_{k+1}$ they were unsuccessful.

Exam	MATHEMATICS : Question...16	
Suggested Solutions		Marker's Comments
(i) $A_r \leq B_r$ $\frac{1}{\sqrt{b_n}} A_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}}\right) A_{n-1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}}\right) A_1$ $< \frac{1}{\sqrt{b_n}} B_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}}\right) B_{n-1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}}\right) B_1$ since $A_r < B_r$ for $r=1, 2, 3, \dots, n$ $\therefore \frac{a_1}{\sqrt{b_1}} + \frac{a_2}{\sqrt{b_2}} + \frac{a_3}{\sqrt{b_3}} + \dots + \frac{a_n}{\sqrt{b_n}} \leq \sqrt{b_1} + \sqrt{b_2} + \sqrt{b_3} + \dots + \sqrt{b_n}$ $\text{or } \sum_{r=1}^n \frac{a_r}{\sqrt{b_r}} \leq \sum_{r=1}^n \sqrt{b_r}$		well done
(ii) $a^2 + b^2 - 2ab = (a-b)^2$ $b^2 + c^2 - 2bc = (b-c)^2$ $c^2 + a^2 - 2ca = (c-a)^2$ $\therefore 2(a^2 + b^2 + c^2 - (ab + bc + ca)) = (a-b)^2 + (b-c)^2 + (c-a)^2$ $(a-b), (b-c), (c-a)$ are real numbers since a, b, c positive real number. $\therefore (a-b)^2 \geq 0$ with equality if and only if $a=b$. similarly for $(b-c)^2, (c-a)^2$ Hence if $a^2 + b^2 + c^2 = ab + bc + ca$ then $(a-b)^2 + (b-c)^2 + (c-a)^2 = 0$ $\therefore (a-b)^2 = (b-c)^2 = (c-a)^2 = 0$ $\therefore a=b \quad b=c \quad c=a$ Hence $a=b=c$ and $\triangle ABC$ is equilateral		students must use $a^2 + b^2 + c^2 = ab + bc + ca$ $\Rightarrow$ then and not assume equality and see if it works.